

Supplementary Material

Towards Balanced Bias–Variance with SincDualFormer: A Dual-Scale Sinc-Filterbank Transformer Model with SR-BandMix Augmentation for Motor Imagery BCI

S1 Full Pseudocode of SincDualFormer and SR-BandMix

This section provides complete pseudocode for the forward procedure of SincDualFormer and the training-pool implementation of SR-BandMix. Algorithm S1 summarizes the end-to-end model path, including the rhythm-specific Sinc and temporal-convolution paths, independent BandSE recalibration and spatial projection, post-spatial fusion, Inception-TCN processing, and Transformer-based classification. Algorithm S2 describes the construction of persistent same-class training pools and the generation of temporally and spectrally augmented samples.

Algorithm S1 Forward procedure of SincDualFormer

Input: Standardized EEG batch $\mathbf{X} \in \mathbb{R}^{B \times 1 \times C \times T}$

Output: Class logits $\hat{\mathbf{y}}$

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1: function RHYTHMBRANCH( $\mathbf{X}, r$ )
2:   if  $r = \mu$  then
3:      $(F_{\min}, F_{\max}) \leftarrow (8, 13)$  Hz
4:      $(K_r, d_r) \leftarrow (K_L, d_L)$ 
5:   else
6:      $(F_{\min}, F_{\max}) \leftarrow (13, 30)$  Hz
7:      $(K_r, d_r) \leftarrow (K_S, d_S)$ 
8:   end if
9:    $\mathbf{Y}_c \leftarrow \text{TempConv}_{K_r, d_r}(\mathbf{X})$   $\triangleright$  data-adaptive temporal path
10:   $\mathbf{Y}_s \leftarrow \text{SincFB}_{[F_{\min}, F_{\max}]}(\mathbf{X})$   $\triangleright$  rhythm-constrained Sinc path
11:   $\mathbf{Y}_c \leftarrow \text{BandSE}_c^{(r)}(\mathbf{Y}_c)$ 
12:   $\mathbf{Y}_s \leftarrow \text{BandSE}_s^{(r)}(\mathbf{Y}_s)$ 
13:   $\mathbf{S}_c \leftarrow \text{DWSpatial}_c^{(r)}(\text{BN}_c^{(r)}(\mathbf{Y}_c))$ 
14:   $\mathbf{S}_s \leftarrow \text{DWSpatial}_s^{(r)}(\text{BN}_s^{(r)}(\mathbf{Y}_s))$ 
15:   $\mathbf{S}^{(r)} \leftarrow \text{PWConv}^{(r)}(\text{Concat}_{\text{ch}}(\mathbf{S}_c, \mathbf{S}_s))$   $\triangleright$  post-spatial fusion
16:   $\mathbf{Z}^{(r)} \leftarrow \text{Dropout}(\text{Pool}_{P_1}(\text{ELU}(\text{BN}(\mathbf{S}^{(r)}))))$ 
17:   $\mathbf{V}^{(r)} \leftarrow \text{ELU}(\text{BN}(\text{ITCN}_{(8,1),(16,2),(32,3)}(\mathbf{Z}^{(r)})))$ 
18:   $\mathbf{H}^{(r)} \leftarrow \text{Reshape}_{\text{token}}(\text{Dropout}(\text{Pool}_{P_2}(\mathbf{V}^{(r)})))$ 
19:  return  $\mathbf{H}^{(r)}$ 
20: end function

21:  $\mathbf{H}^{(\mu)} \leftarrow \text{RHYTHMBRANCH}(\mathbf{X}, \mu)$ 
22:  $\mathbf{H}^{(\beta)} \leftarrow \text{RHYTHMBRANCH}(\mathbf{X}, \beta)$ 
23:  $\mathbf{H} \leftarrow \text{Concat}_{\text{emb}}(\mathbf{H}^{(\mu)}, \mathbf{H}^{(\beta)})$ 
24:  $\mathbf{H}_0 \leftarrow \text{Dropout}_{\text{pos}}(\sqrt{E} \mathbf{H} + \mathbf{P}_{\text{pos}})$ 
25:  $\mathbf{H}_L \leftarrow \text{TransEnc}(\mathbf{H}_0)$ 
26:  $\mathbf{H}_{\text{res}} \leftarrow \mathbf{H}_0 + \mathbf{H}_L$ 
27:  $\hat{\mathbf{y}} \leftarrow \text{Linear}(\text{Dropout}(\text{Flatten}(\mathbf{H}_{\text{res}})))$ 
28: return  $\hat{\mathbf{y}}$ 

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Algorithm S2 Training-pool SR-BandMix augmentation

Input: Official training set $\mathcal{D}_{\text{tr}}$; number of classes K ; mini-batch size B ; augmentation factor N_{aug} ; number of temporal segments M ; sampling rate f_s

Output: Augmented set $\mathcal{D}_{\text{aug}}$

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1: Define  $\mathcal{B} \leftarrow \{[8, 12), [12, 16), [16, 20), [20, 24), [24, 30)\}$  Hz
2: for  $c \leftarrow 0$  to  $K - 1$  do
3:    $\mathcal{P}_c \leftarrow \{\mathbf{x}_i \mid (\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{tr}}, y_i = c\}$   $\triangleright$  persistent training-only pool
4: end for
5:  $n_c \leftarrow N_{\text{aug}} \max(1, \lfloor B/K \rfloor)$ 
6:  $\mathcal{D}_{\text{aug}} \leftarrow \emptyset$ 
7: for  $c \leftarrow 0$  to  $K - 1$  do
8:    $\mathbf{A} \leftarrow \text{SR-RECOMPOSE}(\mathcal{P}_c, n_c, M)$ 
9:    $\mathbf{B} \leftarrow \text{SR-RECOMPOSE}(\mathcal{P}_c, n_c, M)$ 
10:   $\mathbf{Z}_A \leftarrow \text{rDFT}(\mathbf{A})$ 
11:   $\mathbf{Z}_B \leftarrow \text{rDFT}(\mathbf{B})$ 
12:   $\hat{\mathbf{Z}} \leftarrow \mathbf{Z}_A$   $\triangleright$  retain frequencies outside 8–30 Hz
13:  for all  $\mathcal{B}_q \in \mathcal{B}$  do
14:     $\mathcal{K}_q \leftarrow \{k \mid f_k \in \mathcal{B}_q\}$ 
15:    Draw  $\mathbf{g}_q \sim \text{Bernoulli}(0.5)^{n_c}$ 
16:    for  $j \leftarrow 1$  to  $n_c$  do
17:      if  $g_{q,j} = 0$  then
18:         $\mathbf{Z}_j[:, \mathcal{K}_q] \leftarrow \mathbf{Z}_{B,j}[:, \mathcal{K}_q]$ 
19:      end if
20:    end for
21:  end for
22:   $\mathbf{X}_{\text{aug}} \leftarrow \text{irDFT}(\hat{\mathbf{Z}}, T)$ 
23:  Add  $\{(\mathbf{X}_{\text{aug},j}, c)\}_{j=1}^{n_c}$  to  $\mathcal{D}_{\text{aug}}$ 
24: end for
25: Randomly shuffle  $\mathcal{D}_{\text{aug}}$ 
26: return  $\mathcal{D}_{\text{aug}}$ 

27: function SR-RECOMPOSE( $\mathcal{P}, n, M$ )
28:   Divide the trial into  $M$  aligned, non-overlapping intervals  $\{\mathcal{I}_m\}_{m=1}^M$ 
29:   for  $j \leftarrow 1$  to  $n$  do
30:     for  $m \leftarrow 1$  to  $M$  do
31:       Sample  $\mathbf{x} \sim \text{Uniform}(\mathcal{P})$ 
32:        $\mathbf{X}'_j[:, \mathcal{I}_m] \leftarrow \mathbf{x}[:, \mathcal{I}_m]$ 
33:     end for
34:   end for
35:   return  $\mathbf{X}'$ 
36: end function
```

S2 Signal-Level Visualization Before and After Augmentation

To provide an intuitive assessment of how SR-BandMix modifies the EEG signals, we present signal-level before-and-after visualizations for Dataset II. Dataset II was selected because its three-channel configuration, consisting of C3, Cz, and C4, enables the temporal and spectral effects of the augmentation procedure to be displayed clearly. The visualization complements the quantitative augmentation comparisons reported in the main manuscript; it is intended as qualitative evidence of controlled signal variation rather than a formal proof of label preservation or classification improvement.

S2.1 Visualization Protocol

The visualization was conducted using Subject B01 from Dataset II. Consistent with the evaluation protocol in the main manuscript, the official training runs B0101T, B0102T, and B0103T were combined. These runs contained 120, 120, and 160 trials, respectively, resulting in 400 training trials in total. The combined training set was class-balanced, with 200 left-hand and 200 right-hand motor-imagery trials. Each trial contained the C3, Cz, and C4 channels sampled at 250 Hz. The 3–7 s motor-imagery interval was retained, yielding an input size of 3×1000 per trial.

All trials were standardized using the global mean and standard deviation computed exclusively from the three official training runs:

$$\mu_{\text{tr}} = 1.29794877 \times 10^{-8}, \quad \sigma_{\text{tr}} = 4.76501873 \times 10^{-6}. \quad (1)$$

No evaluation-session data were used for normalization, source-pool construction, or augmentation.

For each class, all 200 corresponding training trials formed a persistent same-class augmentation pool. For visualization, 64 SR-BandMix samples were generated per class. This value was selected to obtain a stable class-level PSD estimate and is independent of the augmentation factor used during model training. A fixed random seed of 4 was used only to make the displayed examples reproducible.

Each synthetic sample was created in two stages. First, the 1000-point trial was divided into $M = 8$ aligned and non-overlapping intervals, each covering approximately 0.5 s. At every temporal position, a segment was sampled from a randomly selected trial belonging to the same class. The same source segment was used simultaneously for C3, Cz, and C4, thereby avoiding channel-wise independent recombination. This procedure was performed twice to obtain two independently reconstructed trials, **A** and **B**.

Second, **A** and **B** were transformed into the frequency domain using the real-valued DFT. For each of the five sub-bands

$$[8, 12), [12, 16), [16, 20), [20, 24), \text{ and } [24, 30) \text{ Hz}, \quad (2)$$

the complete complex coefficients were selected from either **A** or **B** with equal probability. The source selection was shared across the three channels within a sub-band. Frequencies outside 8–30 Hz were retained from **A**. The mixed spectrum was then transformed back into the time domain to produce the final SR-BandMix trial.

S2.2 Visualization and Spectral Estimation

Figs. S1 and S2 present the left-hand and right-hand classes, respectively. Each figure contains six complementary panels. Panels (a)–(c) show the standardized C3, Cz, and C4 signals. The traces are vertically offset for readability; therefore, the vertical positions should not be interpreted as absolute amplitudes in microvolts. Panel (a) presents one representative real training trial, panel (b) shows the intermediate same-class Segmentation–Reconstruction sample, and panel (c) shows the final SR-BandMix sample after sub-band selection and inverse transformation.

Panel (d) compares class-level PSDs before and after augmentation. Welch’s method was applied using a 250-sample window, 125-sample overlap, constant detrending, and density scaling. For every trial, the PSD was first averaged across C3, Cz, and C4. The solid curves then show the mean across all trials in the corresponding set: 200 real trials or 64 synthetic trials. The shaded regions represent one standard deviation across trials. The lightly highlighted areas identify the canonical Mu (8–13 Hz) and Beta (13–30 Hz) ranges.

Panels (e) and (f) show the C3 time–frequency representations of the real and augmented examples. Spectrograms were computed with a 128-sample Hann window, 96-sample overlap, a 256-point FFT,

constant detrending, and PSD density scaling. A common display range was obtained from the 5th and 95th percentiles of the combined real and augmented spectrogram values, which allows the two panels to be compared directly. The annotation in panel (f) reports whether each sub-band was inherited from reconstructed sample **A** or **B**. For example, “8–12 Hz:B” indicates that all complex coefficients in that sub-band and across all three channels were taken from **B**.

S2.3 Qualitative Interpretation

For both left- and right-hand imagery, Segmentation–Reconstruction changes the temporal composition of the signal by combining aligned same-class segments, while the final BandMix stage introduces additional frequency-selective variation. The real and augmented PSD curves remain closely aligned across the examined frequency range, including the Mu and Beta regions. This indicates that the augmentation does not cause a large shift in the class-level spectral envelope. At the same time, the curves are not identical, and the waveform and spectrogram panels exhibit localized temporal and spectral differences. These differences illustrate the additional diversity introduced by SR-BandMix rather than simple replication of the original training trials.

The visualizations should be interpreted cautiously. They provide qualitative support that SR-BandMix introduces controlled temporal and spectral variation while broadly maintaining the same-class spectral distribution. They do not, by themselves, prove that every generated trial is artifact-free or that its class label is guaranteed to be preserved. The quantitative effect of the augmentation is therefore assessed separately through the classification, ablation, and augmentation-comparison results reported in the main manuscript.

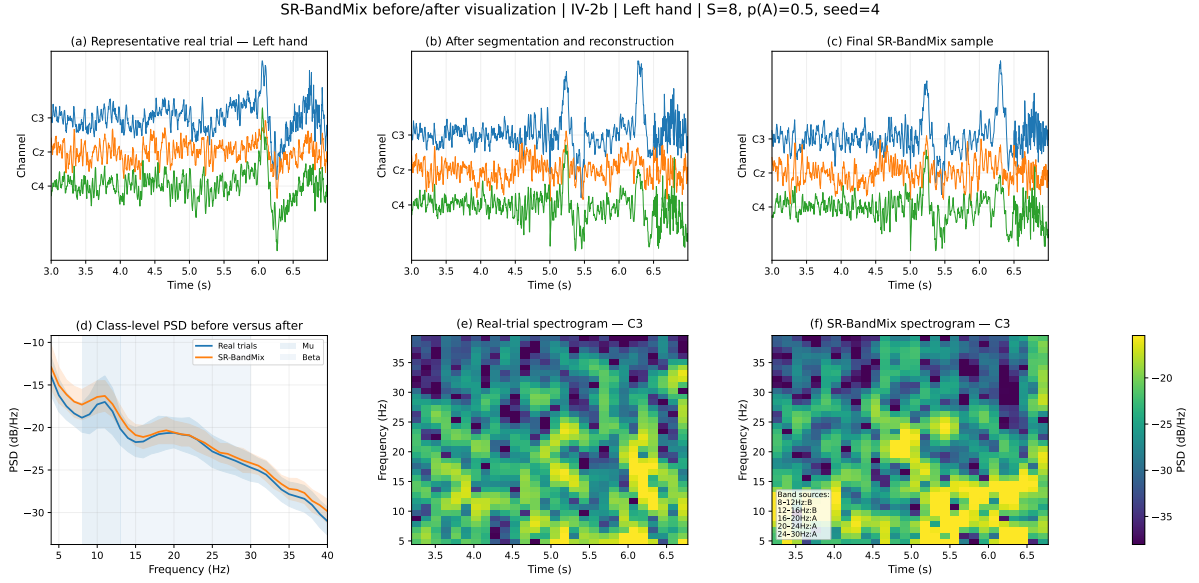


Figure S1: Signal-level visualization of SR-BandMix for the left-hand motor-imagery class of Subject B01 in Dataset II. Panels (a)–(c) show a representative real trial, the intermediate same-class Segmentation–Reconstruction sample, and the final SR-BandMix sample, respectively, across C3, Cz, and C4. Panel (d) compares the class-level PSDs of 200 real training trials and 64 synthetic trials; solid curves denote the means and shaded regions denote one standard deviation across trials. The highlighted regions indicate the Mu and Beta bands. Panels (e) and (f) show the real and augmented C3 spectrograms using a shared display scale. The annotation in panel (f) identifies whether each sub-band was selected from reconstructed sample **A** or **B**.

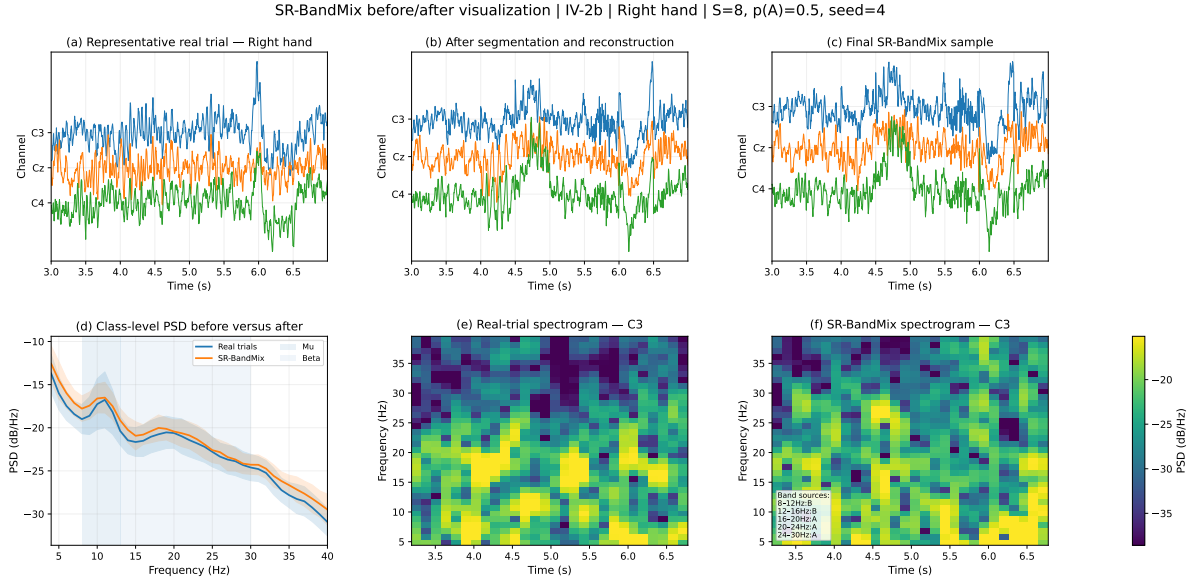


Figure S2: Signal-level visualization of SR-BandMix for the right-hand motor-imagery class of Subject B01 in Dataset II. The visualization follows the same protocol as Fig. S1. The close alignment between the real and augmented class-level PSD curves indicates that the overall spectral envelope is broadly maintained, whereas the waveform and spectrogram differences illustrate the temporal and localized spectral diversity introduced by SR-BandMix.

S3 Subject-Level Statistical Validation of SR-BandMix

This section provides the detailed statistical analysis of the four augmentation conditions: no augmentation, Segmentation–Reconstruction (SR) alone, BandMix alone, and SR-BandMix. The analysis was conducted separately on Datasets I and II.

For all four conditions, the dataset partitions, preprocessing pipeline, SincDualFormer architecture, optimization settings, augmentation factor, and five predefined seeds (0, 1, 2, 3, and 4) were held fixed. Only the augmentation strategy was varied.

To avoid treating repeated stochastic runs as independent observations, the five seed-specific results were first averaged within each subject. The resulting nine paired subject-level observations were used as the experimental units. Dataset-level means and standard deviations in Table S1 were consequently calculated across the nine subject-level means rather than across the 45 subject–seed runs.

For each dataset, a Friedman test was first applied across the four augmentation conditions. Following a significant omnibus test, two-sided Wilcoxon signed-rank tests compared SR-BandMix with no augmentation, SR alone, and BandMix alone. The three post-hoc comparisons within each dataset were corrected using the Holm procedure. Kendall’s W was used as the omnibus effect-size measure, while the matched-pairs rank-biserial correlation, r_{rb} , was reported for each pairwise comparison. A positive r_{rb} indicates an advantage for SR-BandMix.

Table S1: Subject-level descriptive results for the four augmentation conditions. Each subject-level value was first averaged across the five predefined seeds. Mean and SD were then calculated across the nine subjects.

Dataset	Augmentation	Accuracy (%)	Kappa
Dataset I	No augmentation	75.24 $\pm$ 7.72	0.670 $\pm$ 0.103
	SR	81.31 $\pm$ 8.89	0.751 $\pm$ 0.119
	BandMix	77.18 $\pm$ 10.05	0.696 $\pm$ 0.134
	SR-BandMix	82.14 $\pm$ 8.57	0.762 $\pm$ 0.114
Dataset II	No augmentation	86.34 $\pm$ 8.81	0.727 $\pm$ 0.176
	SR	87.53 $\pm$ 9.61	0.751 $\pm$ 0.192
	BandMix	86.70 $\pm$ 8.66	0.734 $\pm$ 0.173
	SR-BandMix	87.93 $\pm$ 9.10	0.759 $\pm$ 0.182

Table S2: Subject-level statistical comparisons of the four augmentation conditions. Pairwise differences are expressed in percentage points (pp) as SR-BandMix minus the comparator. The three pairwise tests within each dataset were adjusted using the Holm procedure.

Friedman omnibus tests					
Dataset	$\chi^2(3)$	p -value	Kendall’s W	Conclusion	
Dataset I	22.733	< 0.001	0.842	Significant	
Dataset II	10.200	0.017	0.378	Significant	

Post-hoc Wilcoxon signed-rank tests					
Dataset	Comparator	Δ Acc. (pp)	r_{rb}	Raw p	Holm p
Dataset I	No augmentation	6.90	1.000	0.004	0.012
	SR	0.83	0.711	0.059	0.059
	BandMix	4.95	1.000	0.004	0.012
Dataset II	No augmentation	1.59	0.778	0.039	0.117
	SR	0.40	0.244	0.547	0.547
	BandMix	1.24	0.556	0.156	0.313

The omnibus analysis shows a significant difference among the four augmentation conditions on both

datasets. On Dataset I, SR-BandMix significantly outperformed no augmentation and BandMix alone after Holm correction. Its improvement over SR alone did not reach statistical significance. On Dataset II, SR-BandMix achieved the highest mean accuracy, but none of its pairwise advantages remained significant after Holm correction.

We additionally evaluated the interaction contrast

$$\Delta_{\text{int}} = (\text{SR-BandMix} - \text{SR}) - (\text{BandMix} - \text{NoAug}). \quad (3)$$

The interaction was not significant on Dataset I (mean interaction = -1.12 pp, $p = 0.461$) or Dataset II (mean interaction = 0.04 pp, $p = 0.984$). Thus, the present results do not establish a statistically significant synergistic interaction between SR and BandMix.

Overall, SR-BandMix achieved the highest mean accuracy on both public datasets, but its incremental contribution beyond SR alone was modest and dataset-dependent. SR provided the more consistent independent improvement, whereas the additional contribution of BandMix was clearer on Dataset I than on Dataset II. These results support a moderated interpretation of SR-BandMix rather than a claim of uniform statistical superiority across all augmentation conditions.